Experiment No: 4 - Indexing and Slicing of NumPy Arrays

Aim

To study indexing and slicing operations on 1-D, 2-D, and 3-D NumPy arrays, including negative slicing.

Software Required

Python 3.x, NumPy Library, Jupyter Notebook / Google Colab / Python IDLE

Theory

NumPy supports efficient indexing and slicing to access subsets of array data.

a) Slicing 1-D NumPy Arrays

Program:

```
import numpy as np
arr=np.array([10,20,30,40,50,60])
print(arr[1:5])
```

Output:

```
[20 30 40 50]
```

b) Slicing 2-D NumPy Arrays

Program:

```
import numpy as np
arr=np.array([[1,2,3],[4,5,6],[7,8,9]])
print(arr[0:2,1:3])
```

Output:

```
[[2 3]
 [5 6]]
```

c) Slicing 3-D NumPy Arrays

Program:

```
import numpy as np
arr=np.array([[[1,2],[3,4]],[[5,6],[7,8]]])
print(arr[0,:,0:2])
```

Output:

```
[[1 2]
 [3 4]]
```

d) Negative Slicing of NumPy Arrays

Program:

```
import numpy as np
arr=np.array([10,20,30,40,50,60])
print(arr[-5:-1])
```

Output:

[20 30 40 50]

Result

Thus, indexing and slicing operations on 1-D, 2-D, and 3-D NumPy arrays, including negative slicing, were successfully performed.